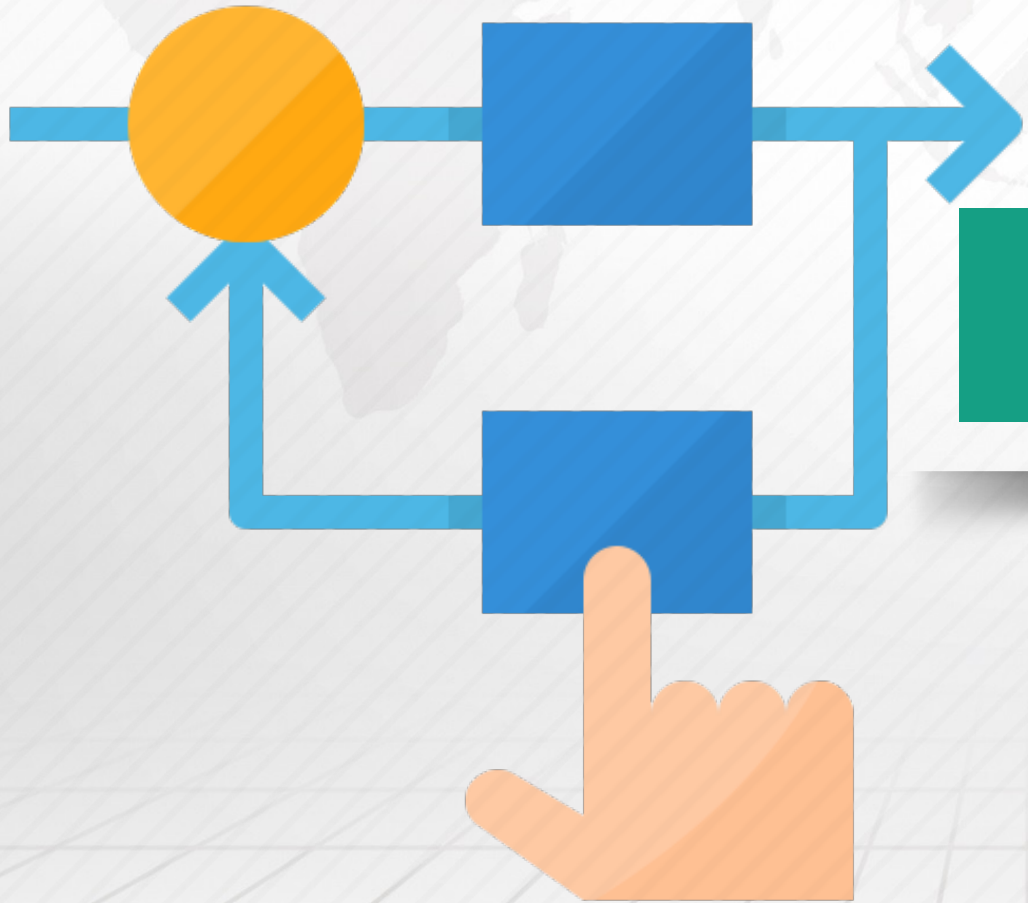




江苏科技大学
JIANGSU UNIVERSITY OF SCIENCE AND TECHNOLOGY



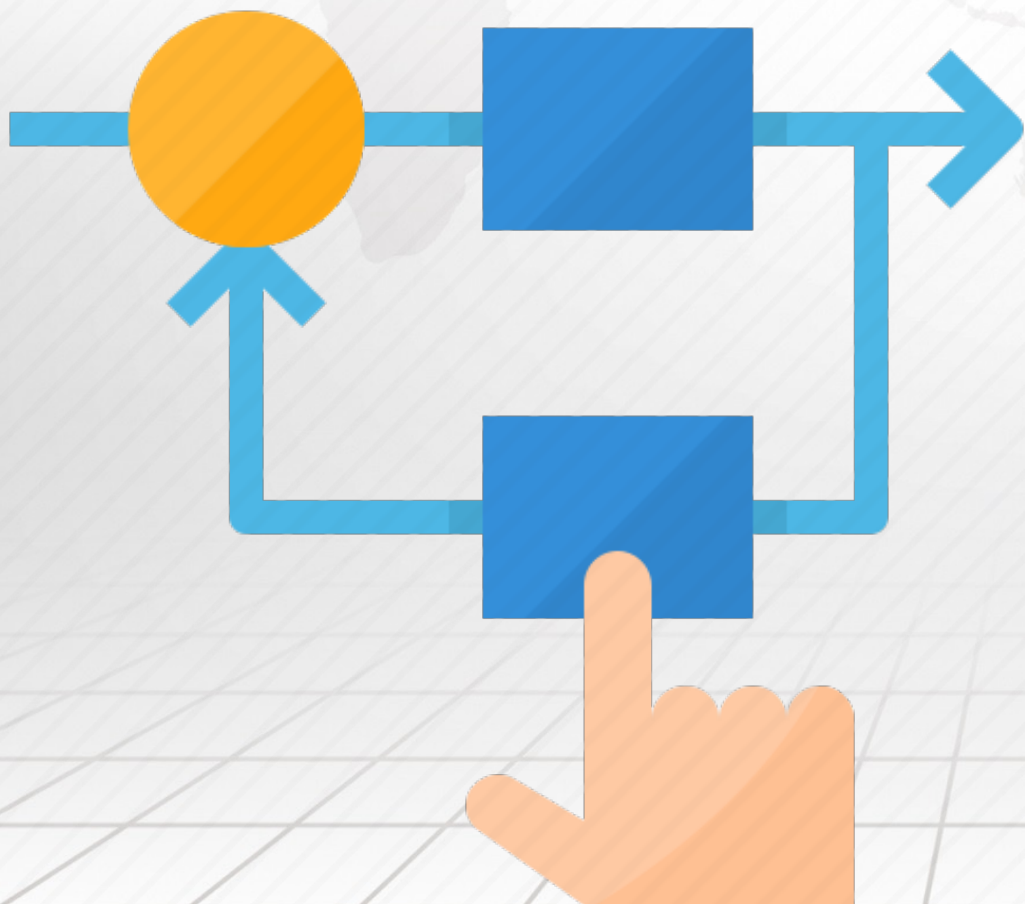
自动控制原理

第一章	第二章	第三章	第四章	第五章	第六章
基本概念	数学模型	时域分析法	根轨迹法	频域分析法	校正方法

15. 根轨迹法：图形化思考



学习目标



01

能熟练绘制根轨迹

重

能基于根轨迹分析参数稳定范围

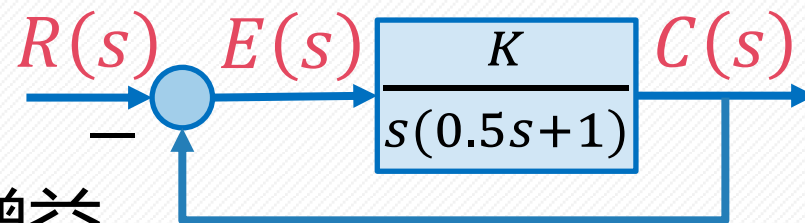
03

能够将根轨迹增益转化为开环增益

难

能够使用根轨迹选择满足性能的参数

利用根轨迹分析系统性能例1



$$D(s) = s^2 + 2s + K^* = 0 \quad \begin{cases} K: & \text{开环增益} \\ K^* = 2K: & \text{根轨迹增益} \end{cases}$$

$$\lambda_{1,2} = -1 \pm \sqrt{1 - K^*}$$

$$K^* = 2K = 0 \xrightarrow{\zeta > 1} 1 \xrightarrow{0 < \zeta < 1} \infty$$

动态性能

$$\sigma\% = 0 \quad \beta \uparrow \Rightarrow \zeta \downarrow \Rightarrow \sigma\% \uparrow$$

$$t_s \downarrow \quad t_s = \frac{3.5}{\zeta \omega_n}$$

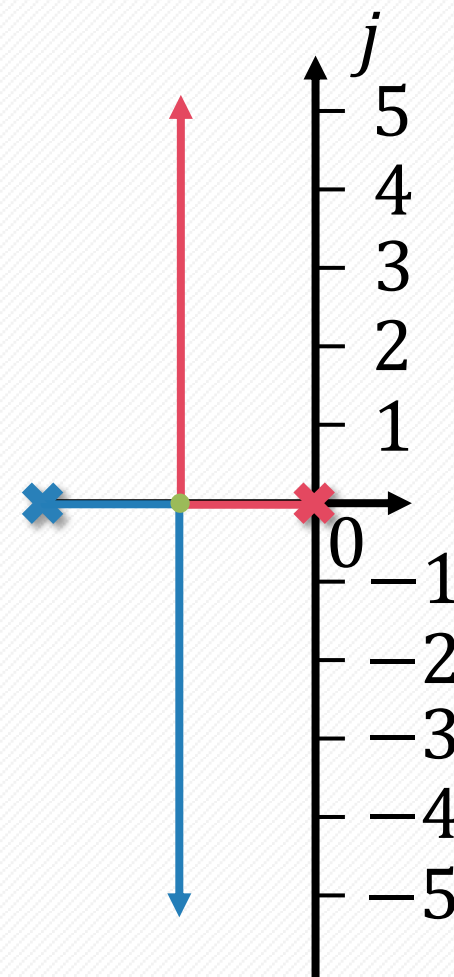
稳定性

$$\text{Re}[\lambda_{1,2}] < 0, \text{ 系统绝对稳定}$$

稳态误差

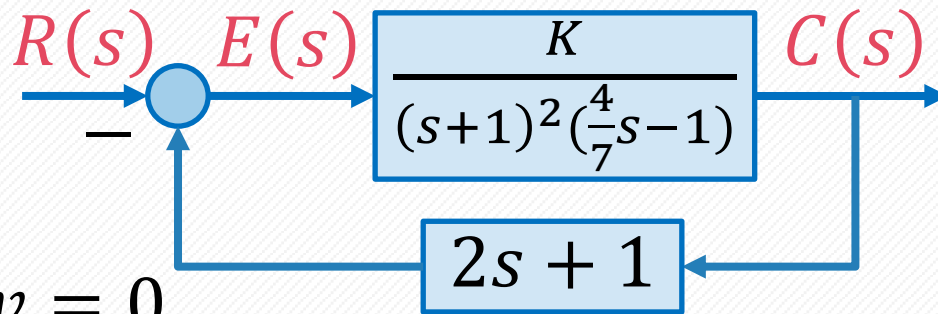
$$r(t) = At$$

$$K^* \uparrow \Rightarrow e_{ss} = \frac{A}{K} = \frac{2A}{K^*} \downarrow$$



利用根轨迹分析系统性能例2

$$G(s) = \frac{K(2s+1)}{(s+1)^2(\frac{4}{7}s-1)} = \frac{3.5K(s+\frac{1}{2})}{(s+1)^2(s-\frac{7}{4})} \quad K^* = 3.5K, v = 0$$



01 实轴轨迹

$$[-0.5, 1.75]$$

02 渐近线

$$\sigma = \frac{-2 + 7/4 + 1/2}{3 - 1} = \frac{1}{8}$$

$$\varphi = \frac{(2k+1)\pi}{3-1} = \pm 90^\circ$$

03 出射角

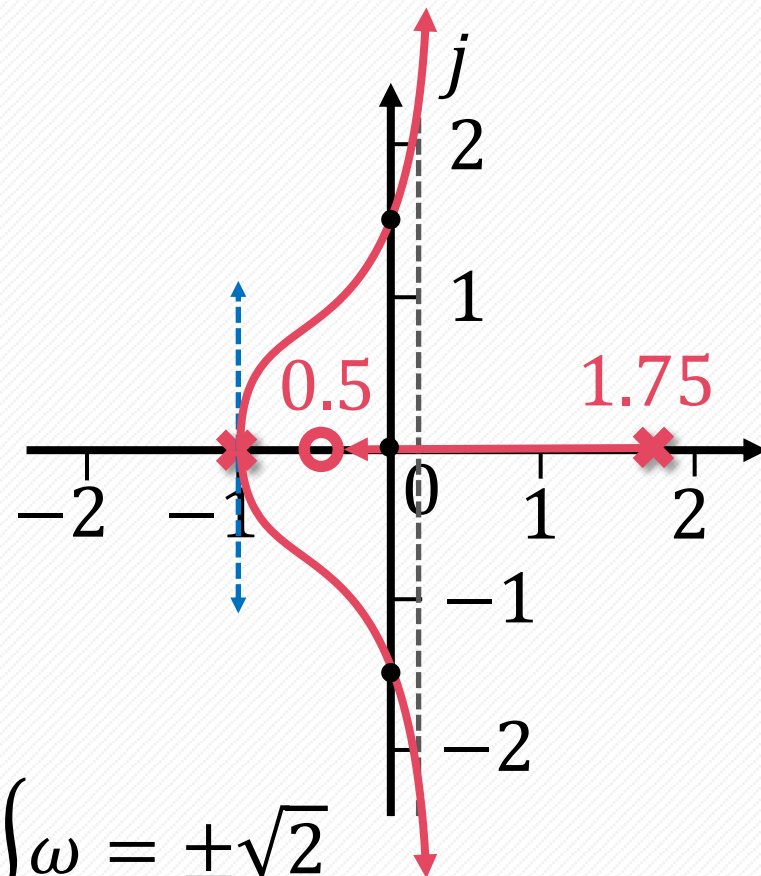
$$\theta = \pm \frac{\pi}{q}$$

04 虚轴交点

$$D(s) = 4s^3 + s^2 + (14K - 10)s + 7(K - 1) = 0$$

$$\begin{cases} \text{Re}[D(j\omega)] = -\omega^2 + 7(K - 1) = 0 \\ \text{Im}[D(j\omega)] = -4\omega^3 + (14K - 10)\omega = 0 \end{cases}$$

$$\begin{cases} \omega = 0 \\ K = 1 \end{cases}, \begin{cases} \omega = \pm\sqrt{2} \\ K = 9/7 \end{cases}$$



利用根轨迹分析系统性能例2

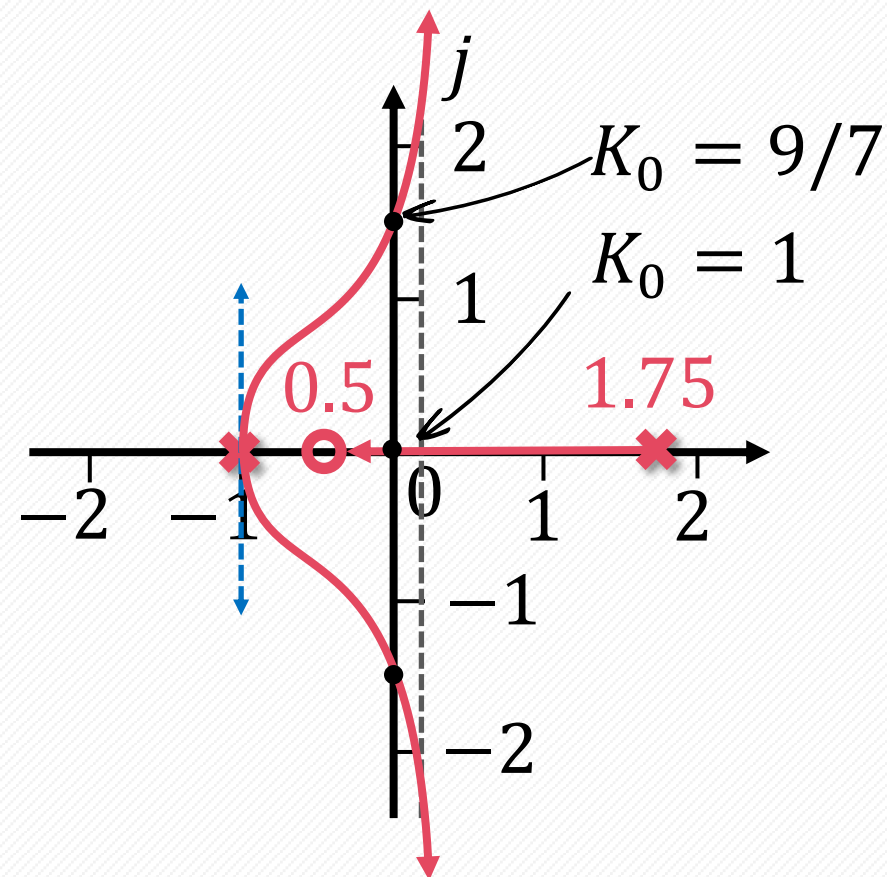
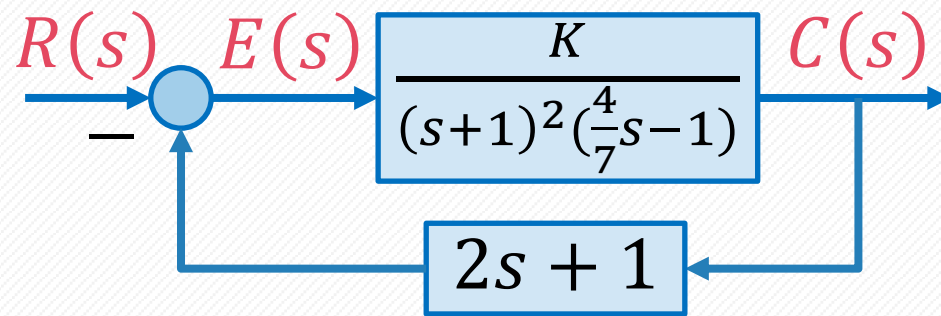
?

系统稳定性如何随 K 变化?

$$\begin{cases} \omega = 0 \\ K = 1 \end{cases}, \begin{cases} \omega = \pm\sqrt{2} \\ K = 9/7 \end{cases}$$

$K = 0 \xrightarrow{\text{不稳定}} K_0 = 1 \xrightarrow{\text{稳定}} K_\omega = \frac{9}{7} \xrightarrow{\text{不稳定}} \infty$

条件稳定



利用根轨迹分析系统性能例3

$$G(s) = \frac{K^*}{s(s+20)(s+2+j4)(s+2-j4)}$$

$$K = K^*/400$$

01 实轴轨迹

$$[-20, 0]$$

02 渐近线

$$\sigma = \frac{0 - 20 - 2 - 2}{4} = -6$$

$$\varphi = \frac{(2k+1)\pi}{4} = \pm 45^\circ, \pm 135^\circ$$

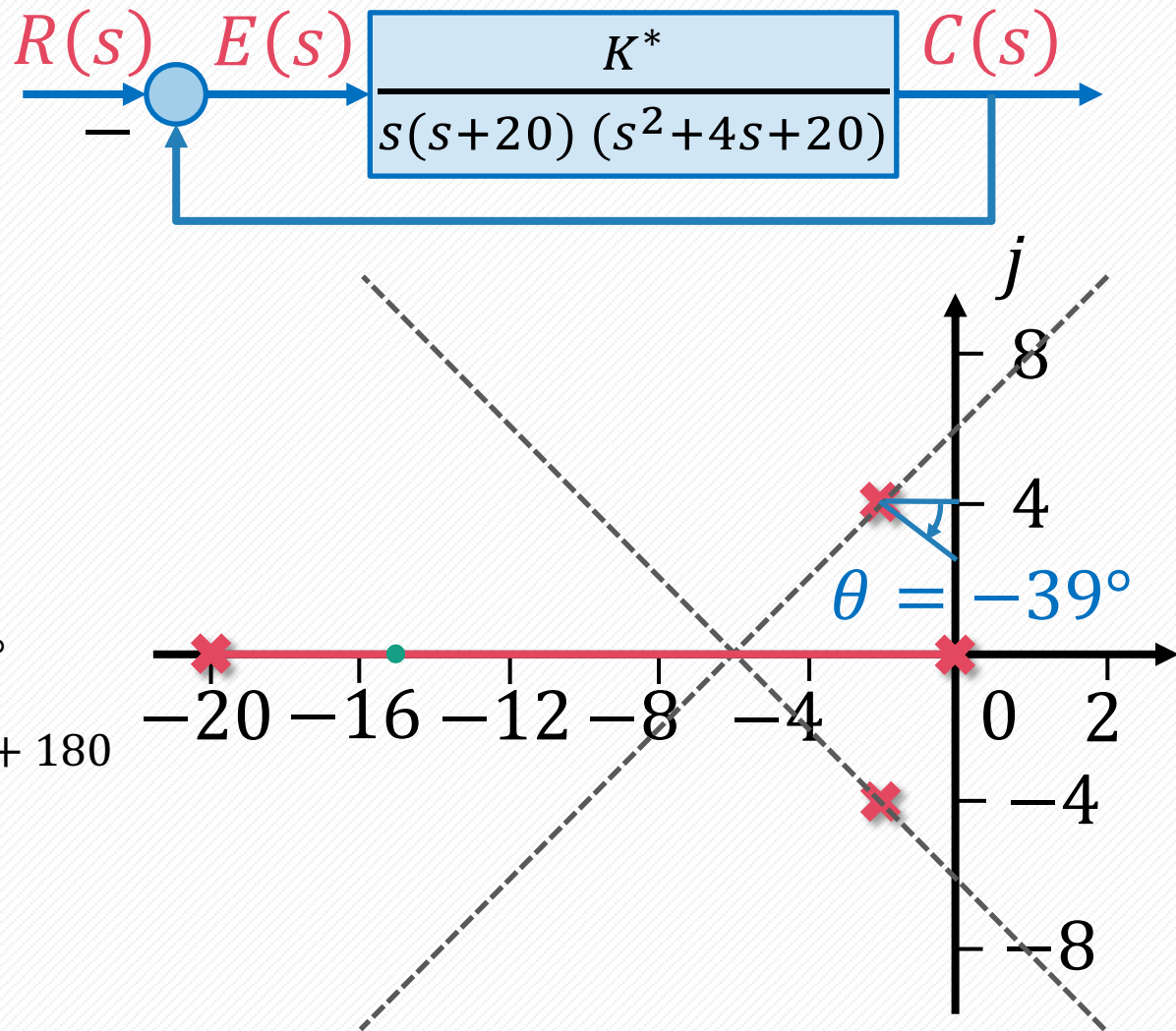
03 出射角

$$\begin{aligned}\theta_1 &= -(90^\circ + 116.5^\circ + 12.5^\circ) + 180^\circ \\ &= -39^\circ\end{aligned}$$

04 分离点

$$\frac{1}{d} + \frac{1}{d+20} + \frac{1}{d+2+j4} + \frac{1}{d+2-j4} = 0 \Rightarrow \frac{1}{d} + \frac{1}{d+20} + \frac{2(d+2)}{(d+2)^2 + 4^2} = 0, \text{ 试根得 } d = -15.1$$

$$K_d^* = |d||d+20||d^2+4d+20| \stackrel{d=-15.1}{=} 13881$$



利用根轨迹分析系统性能例3

$$G(s) = \frac{K^*}{s(s+20)(s+2+j4)(s+2-j4)}$$

$$K = K^*/400$$

01 实轴轨迹

$$[-20, 0]$$

02 渐近线

$$\sigma = \frac{0 - 20 - 2 - 2}{4} = -6$$

$$\varphi = \frac{(2k+1)\pi}{4} = \pm 45^\circ, \pm 135^\circ$$

03 出射角

$$\theta_1 = -(90^\circ + 116.5^\circ + 12.5^\circ) + 180^\circ = -39^\circ$$

04 分离点

$$d = -15.1$$

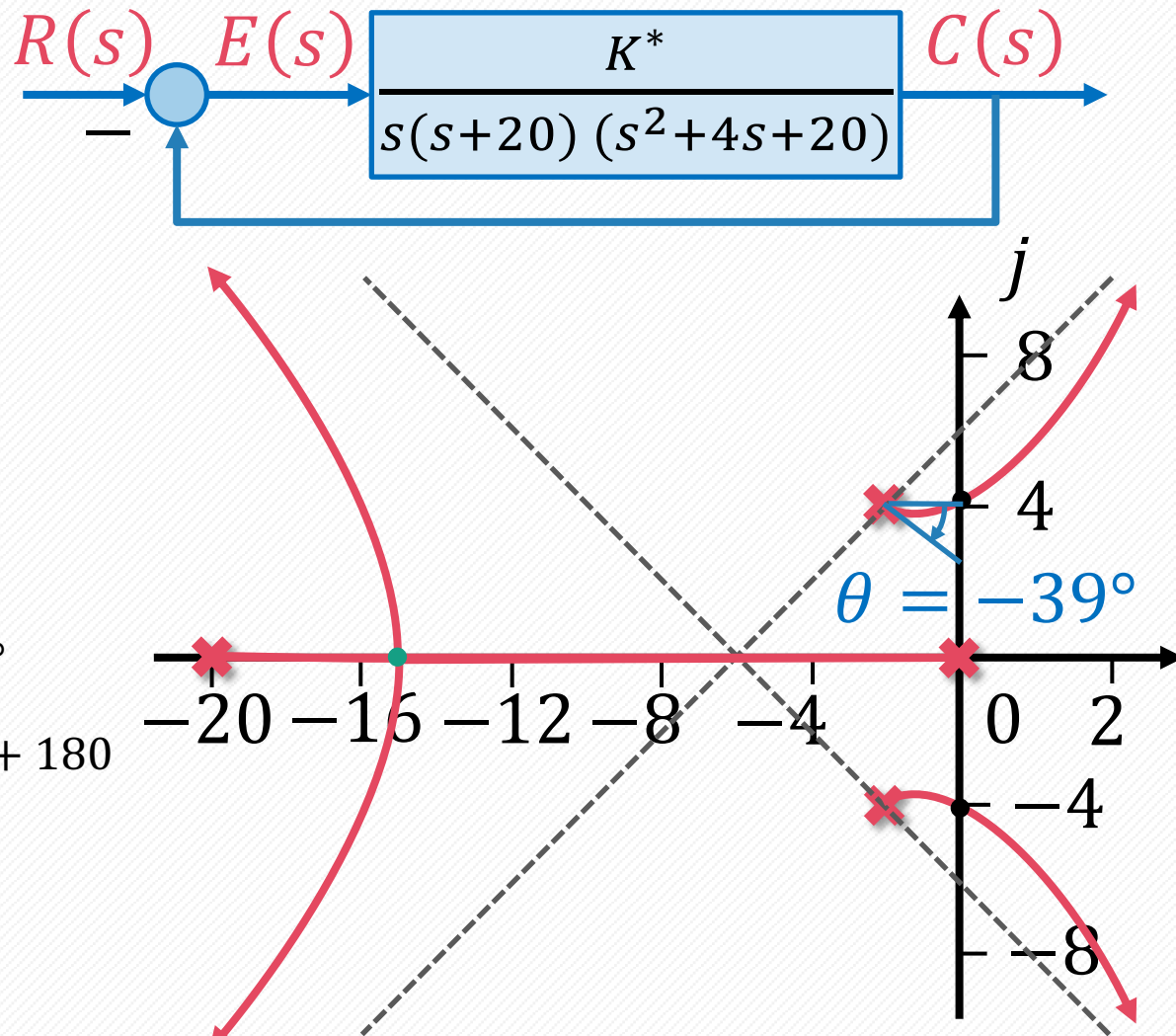
$$K_d^* = 13881$$

05 虚轴交点

$$D(s) = s^4 + 24s^3 + 100s^2 + 400s + K^* = 0$$

$$\begin{cases} \operatorname{Re}[D(j\omega)] = \omega^4 \pm 100\omega^2 + K^* = 0 \\ \operatorname{Im}[D(j\omega)] = -24\omega^3 + 400\omega = 0 \end{cases}$$

$$\begin{cases} \omega = \sqrt{400/24} = 4.1 \\ K^* = 1389 \end{cases}$$



利用根轨迹分析系统性能例3

$$G(s) = \frac{K^*}{s(s+20)(s+2+j4)(s+2-j4)}$$

$$K = K^*/400$$

01 实轴轨迹

$$[-20, 0]$$

02 渐近线

$$\sigma = \frac{0 - 20 - 2 - 2}{4} = -6$$

$$\varphi = \frac{(2k+1)\pi}{4} = \pm 45^\circ, \pm 135^\circ$$

03 出射角

$$\theta_1 = -(90^\circ + 116.5^\circ + 12.5^\circ) + 180^\circ = -39^\circ$$

04 分离点

$$d = -15.1$$

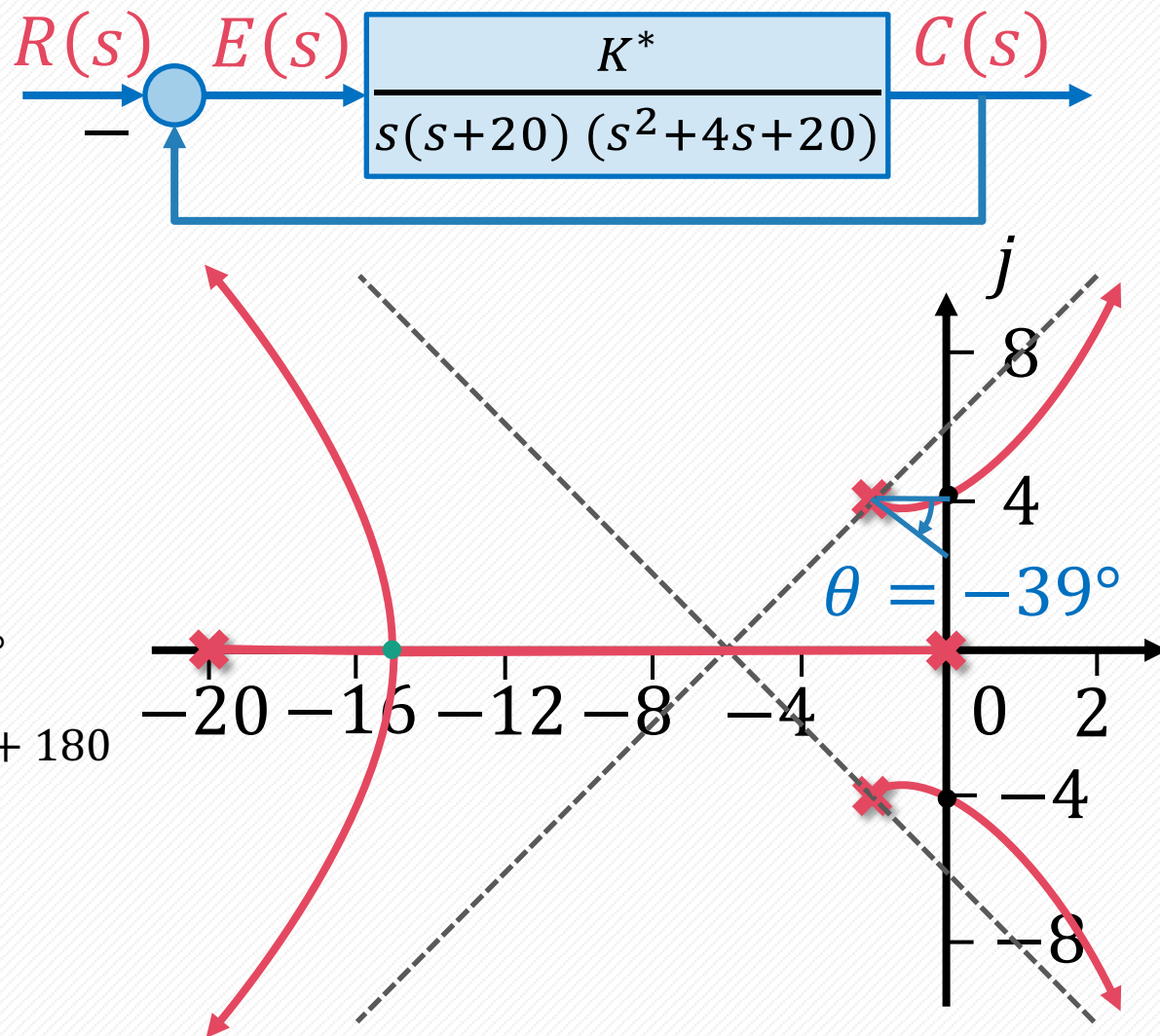
$$K_d^* = 13881$$

05 虚轴交点

$$\begin{cases} \omega = \sqrt{400/24} = 4.1 \\ K^* = 1389 \end{cases}$$

稳定的开环增益范围?

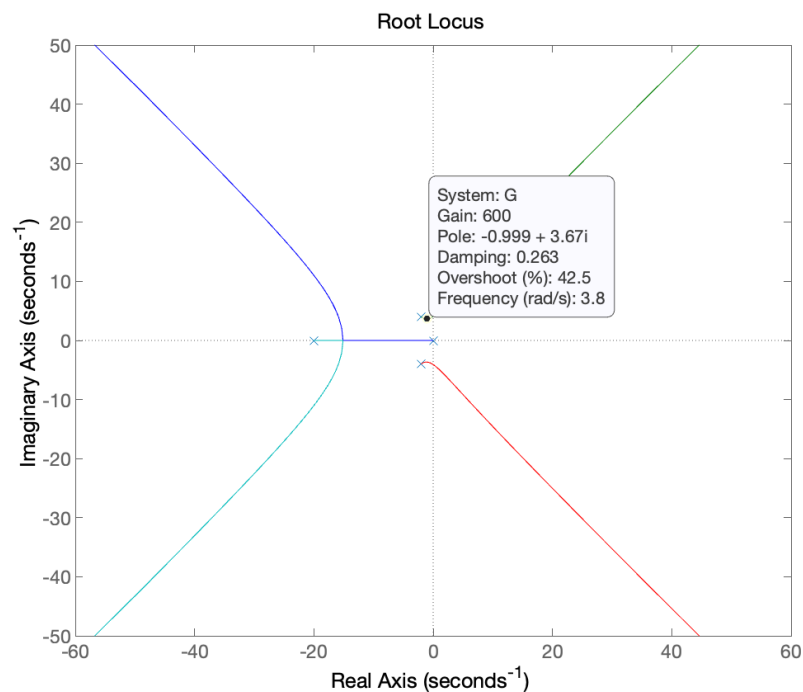
$$0 < K < 3.4725$$



利用根轨迹分析系统性能例3

代码

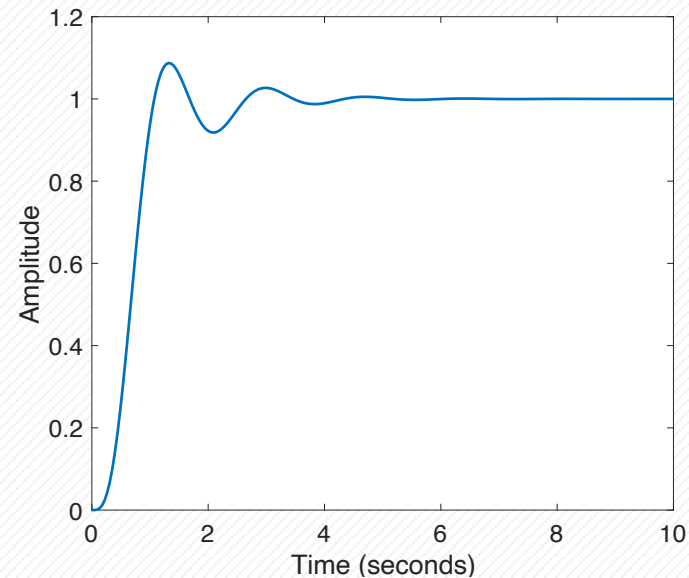
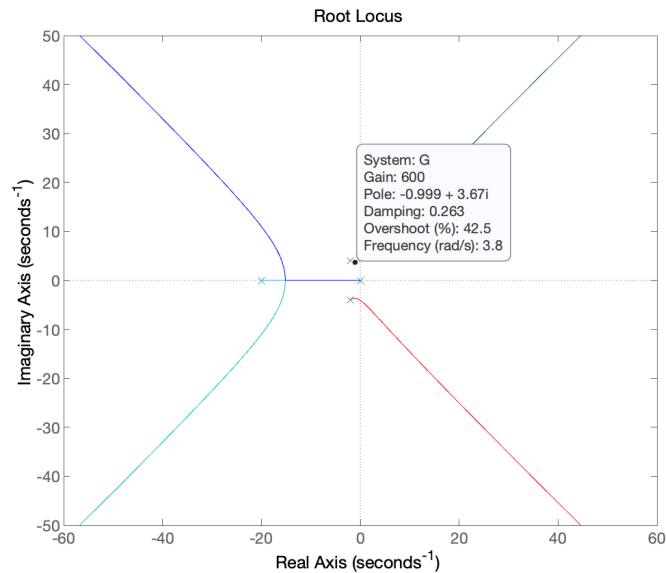
```
G = zpk([], [0, -20, -2-4i, -2+4i], 1); % 零极点法建立开环传函  
rlocus(G) % 绘制根轨迹
```



利用根轨迹分析系统性能例3

代码

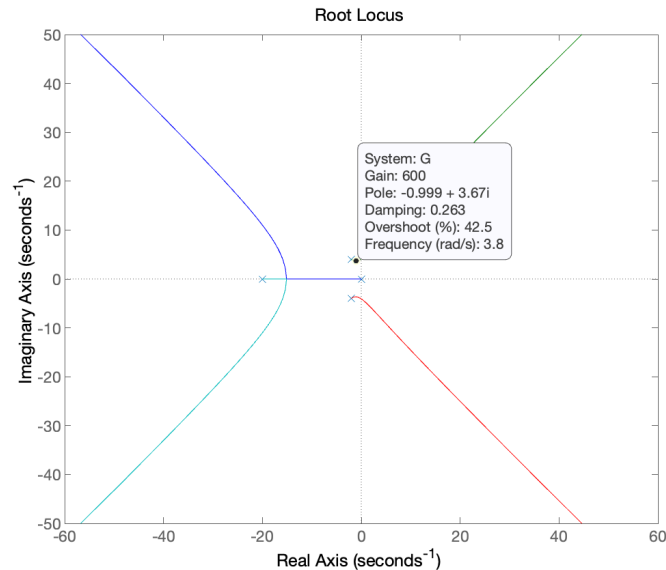
```
G = zpk([], [0, -20, -2-4i, -2+4i], 600); % 以选择的根轨迹增益构建新开环传函
Phi = feedback(G, 1); % 构成单位负反馈系统
[y, t] = step(Phi, 10); % 阶跃响应
plot(t, y, 'LineWidth', 2); % 绘图
xlabel('Time (seconds)')
ylabel('Amplitude')
```



利用根轨迹分析系统性能例3

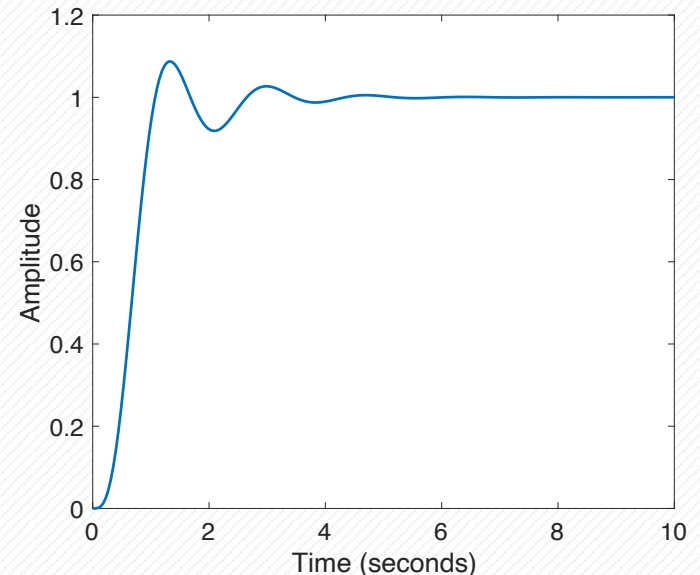
代码

```
G = zpk([], [0, -20, -2-4i, -2+4i], 600); % 以选择的根轨迹增益构建新开环传函
Phi = feedback(G, 1); % 构成单位负反馈系统
[y, t] = step(Phi, 10); % 阶跃响应
plot(t, y, 'LineWidth', 2); % 绘图
xlabel('Time (seconds)')
ylabel('Amplitude')
```



?

为什么选定极点的
阻尼只有0.263，单
位阶跃响应却没有
出现较大的超调呢？



利用根轨迹分析系统性能例4

$$G(s) = \frac{K^*(s+1)}{s(s-1)(s+2+j2\sqrt{3})(s+2-j2\sqrt{3})}$$
$$K = K^*/16$$

01 实轴轨迹

$$(-\infty, -1] \cup [0, 1]$$

02 渐近线

$$\sigma = \frac{1-4+1}{3} = -2/3$$
$$\varphi = \frac{(2k+1)\pi}{3} = \pm 60^\circ, 180^\circ$$

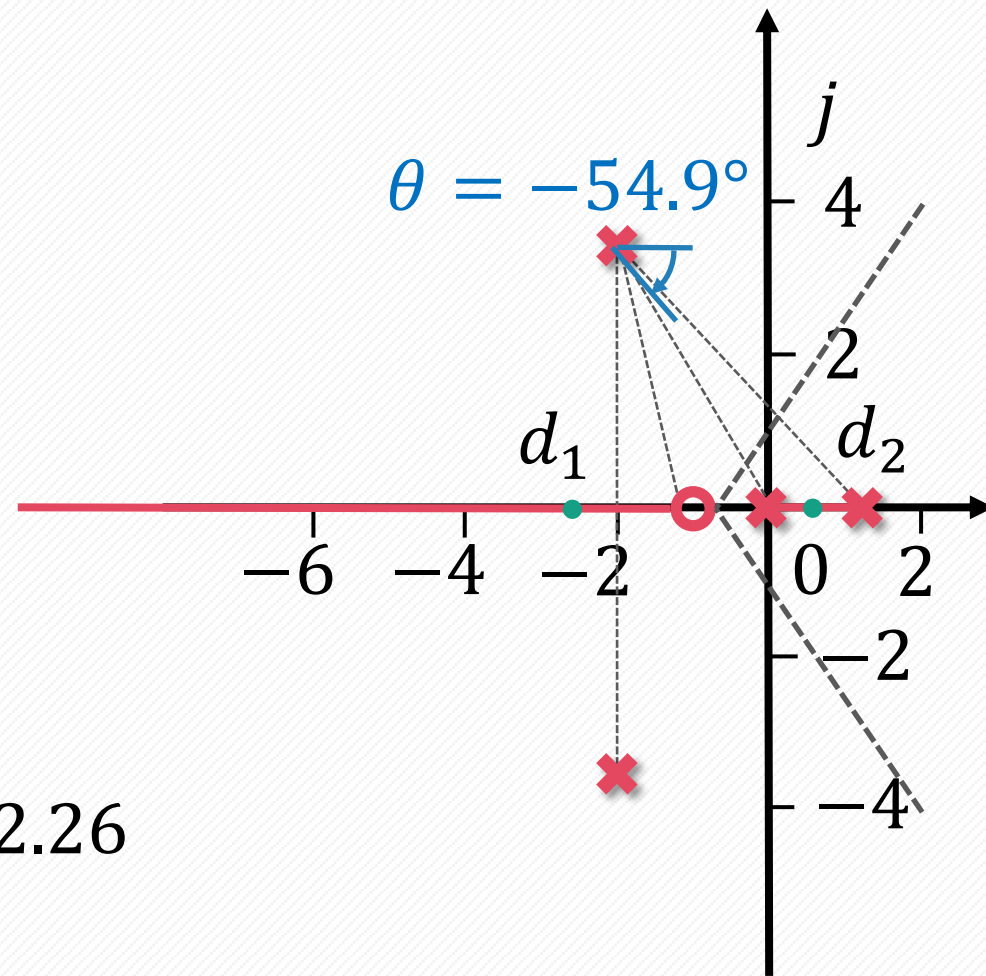
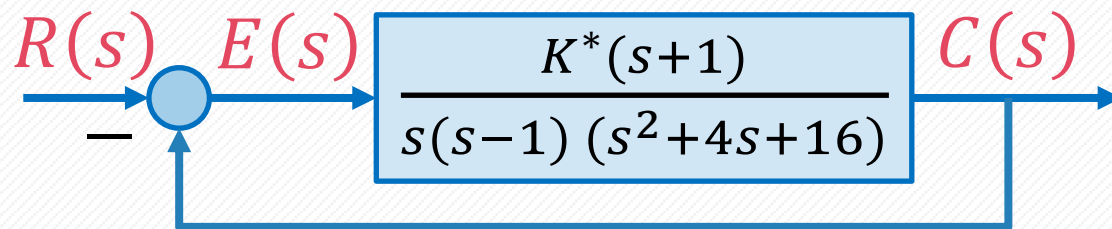
03 出射角

$$\theta_1 = 106.1^\circ - (90^\circ + 120^\circ + 130.9^\circ) + 180 = -54.9^\circ$$

04 分离点

$$\frac{1}{d} + \frac{1}{d-1} + \frac{2(d+2)}{d^2+4d+16} = \frac{1}{d+1} \Rightarrow d_1 = 0.49, d_2 = -2.26$$

$$K_{d_{1,2}}^* = \frac{|d||d-1||d^2+4d+16|}{|d+1|} \Big|_{\substack{d=0.49 \\ d=-2.26}} = 3.05, 70.6$$



利用根轨迹分析系统性能例4

$$K = K^*/16$$

01 实轴轨迹

$$(-\infty, -1] \cup [0, 1]$$

02 渐近线

$$\sigma = \frac{1 - 4 + 1}{3} = -2/3$$

$$\varphi = \frac{(2k+1)\pi}{3} = \pm 60^\circ, \pm 180^\circ$$

03 出射角

$$\theta_1 = -54.9^\circ$$

04 分离点

$$d_1 = 0.49, K_{d_1}^* = 3.05$$

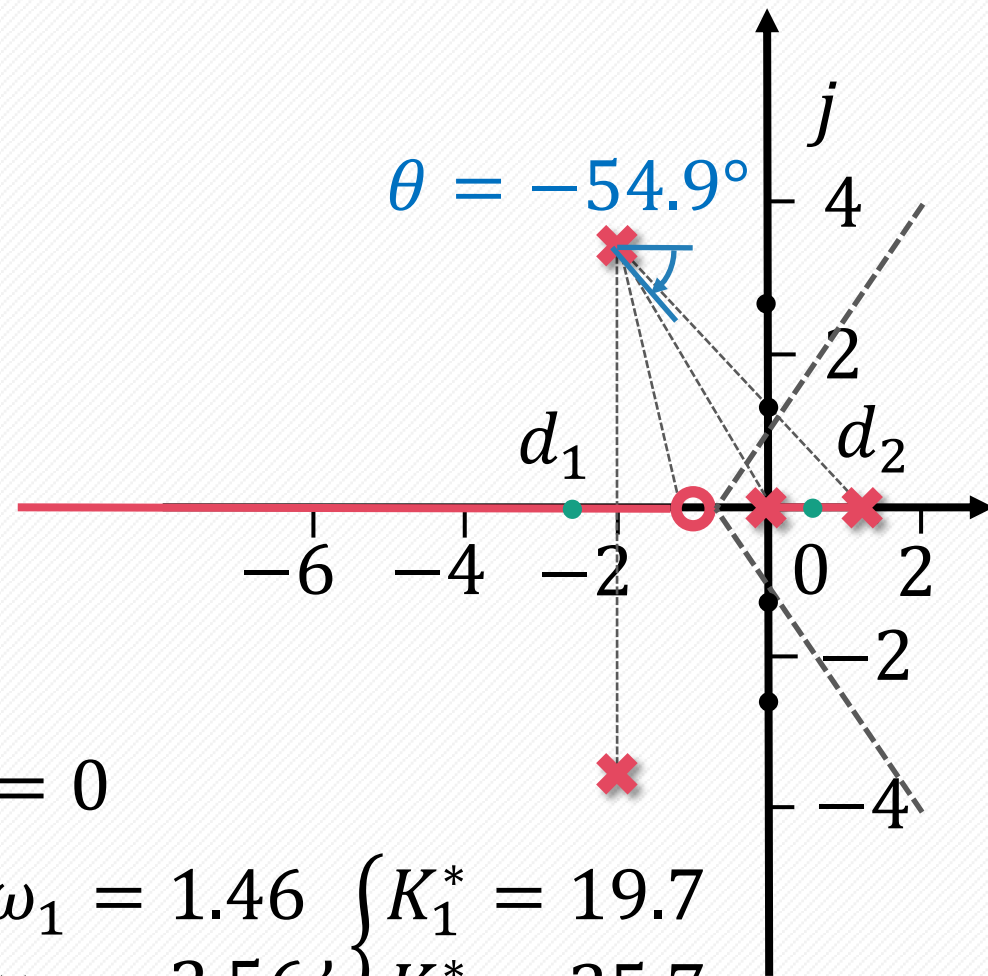
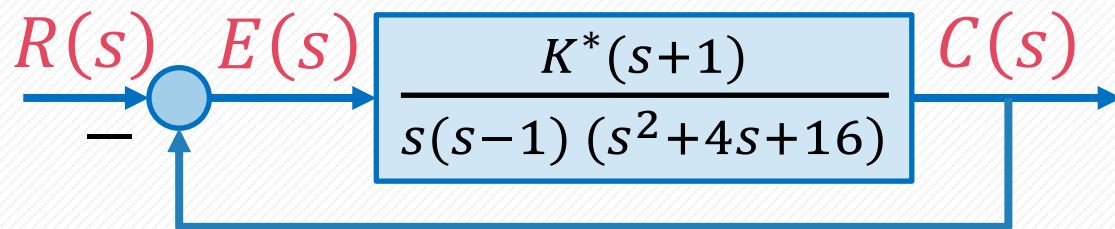
$$d_2 = -2.26, K_{d_2}^* = 70.6$$

05 虚轴交点

$$D(s) = s^4 + 3s^3 + 12s^2 + (K^* - 16)s + K^* = 0$$

$$\begin{cases} \text{Re}[D(j\omega)] = \omega^4 - 12\omega^2 + K^* = 0 \\ \text{Im}[D(j\omega)] = -3\omega^3 + (K^* - 16)\omega = 0 \end{cases}$$

$$\begin{cases} \omega_1 = 1.46 \\ \omega_2 = 2.56 \end{cases} \begin{cases} K_1^* = 19.7 \\ K_2^* = 35.7 \end{cases}$$



利用根轨迹分析系统性能例4

$$K = K^*/16$$

01 实轴轨迹

$$(-\infty, -1] \cup [0, 1]$$

02 渐近线

$$\sigma = \frac{1 - 4 + 1}{3} = -2/3$$

$$\varphi = \frac{(2k+1)\pi}{3} = \pm 60^\circ, \pm 180^\circ$$

03 出射角

$$\theta_1 = -54.9^\circ$$

04 分离点

$$d_1 = 0.49, K_{d_1}^* = 3.05$$

$$d_2 = -2.26, K_{d_2}^* = 70.6$$

05 虚轴交点

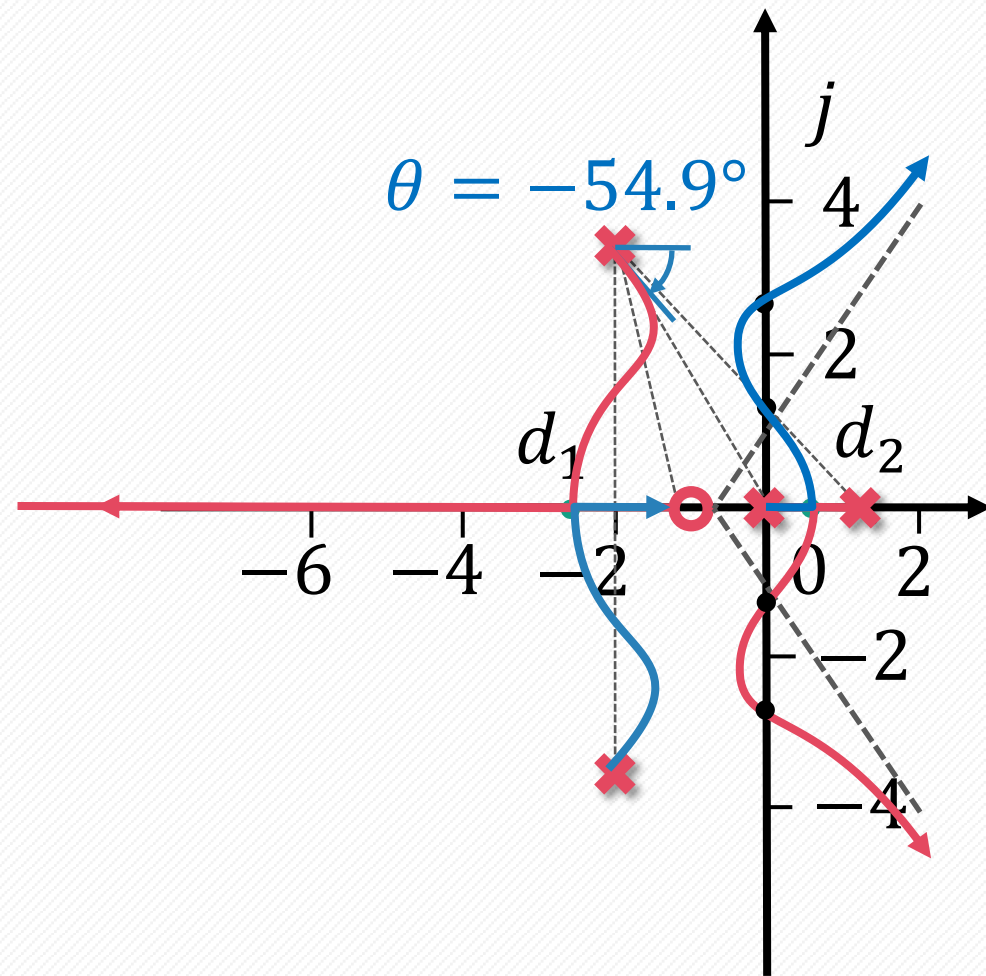
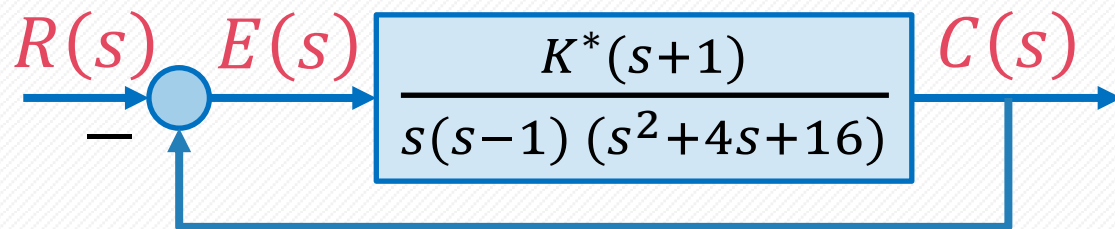
$$\begin{cases} \omega_1 = 1.46 \\ \omega_2 = 2.56 \end{cases} \begin{cases} K_1^* = 19.7 \\ K_2^* = 35.7 \end{cases}$$

稳定的 K^* 范围:

$$19.7 < K^* < 35.7$$

稳定的 K 范围:

$$1.234 < K < 2.23$$



总结

01 根轨迹要点

起点、终点、实轴根轨迹、分离点，出射角，入射角，渐近线，虚轴交点，根之和

02 参数稳定范围

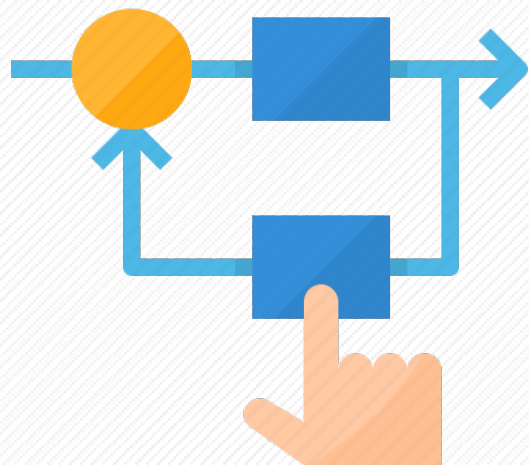
以虚轴为边界

03 根轨迹增益

应用时需化为开环增益

04 选择极点

应用主导极点选择，考虑阻尼和振荡频率



图形化思考

工程思维

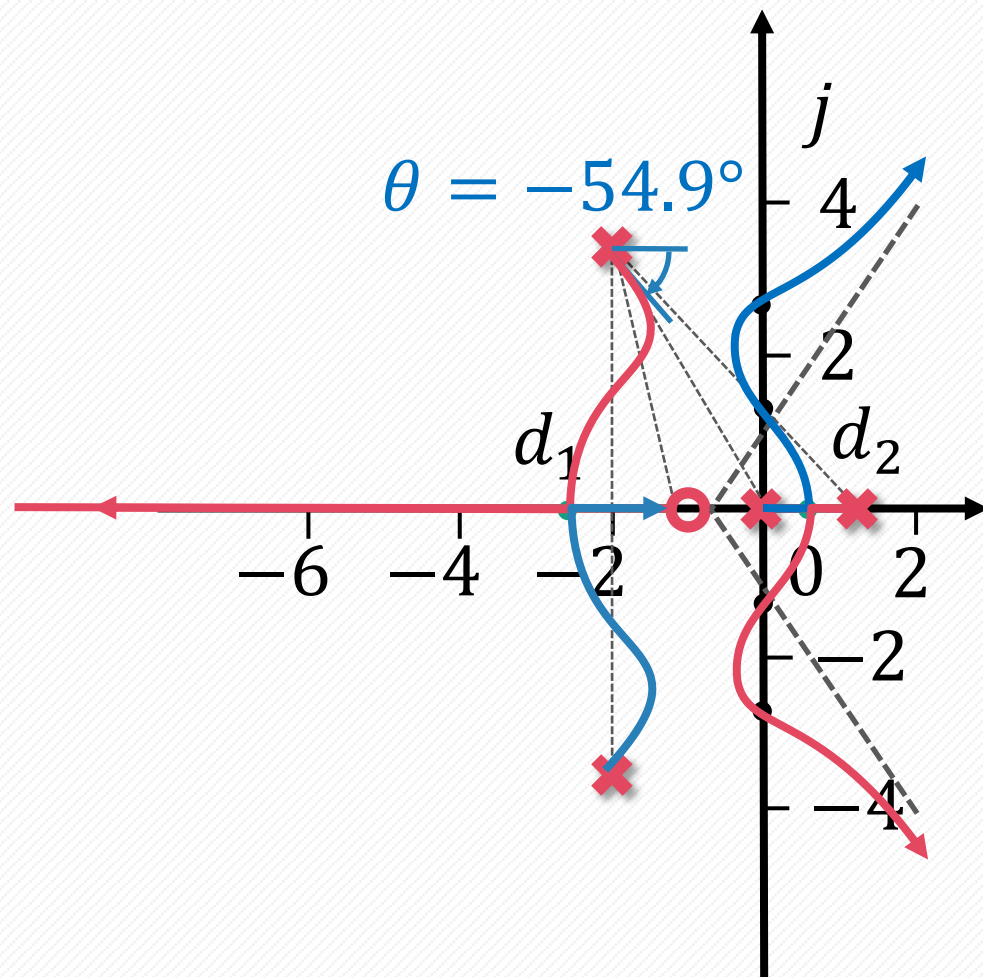
课后思考

?

分析例4中稳定情况下系统性能指标的范围

?

如性能指标不满足要求，考虑可能的校正手段



课外拓展及预习要求



DR_CAN含有初始条件的传递函数:

https://www.bilibili.com/video/BV13J411m7n1?share_source=copy_web

- [1] Doria-Cerezo, A., & Bodson, M. (2016). Design of controllers for electrical power systems using a complex root locus method. IEEE transactions on Industrial Electronics, 63(6), 3706-3716.
- [2] Bollinger, K., Laha, A., Hamilton, R., & Harras, T. (1975). Power stabilizer design using root locus methods. IEEE Transactions on power apparatus and systems, 94(5), 1484-1488.

15. 根轨迹法：图形化思考

感谢聆听！

